

Markov chains & stationary distributions
Markov chain. Finite states S . Future depends only on current state: $\Pr(X_k = j \mid X_{k-1} = i, X_{k-2}, \dots, X_0) = \Pr(X_k = j \mid X_{k-1} = i)$. **Time-homog.** Transition probs do not depend on k : $P_{ij} = \Pr(X_k = j \mid X_{k-1} = i)$, each row sums to 1. For row distribution $p_i(k) = \Pr(X_k = i)$: $p(k+1) = p(k)P$, hence $p(k) = p(0)P^k$.
Stationary distribution. A distribution unchanged by one transition: $w = wP$, $w_i \geq 0$, $\omega_1 = 1$. Compute by solving $w(I - P) = 0$ with $w_i = 1$. **Reachable/irreducible.** j reachable from i if $\exists T$ with $(P^T)_{ij} > 0$. Irreducible means every pair communicates. **Thm 1.** finite irreducible MC $\Rightarrow$ unique stationary w . **Period.** i has period d if return times to i are multiples of d ; equiv. $d(i) = \gcd\{t : (P^t)_{ii} > 0\}$. Aperiodic iff all $d(i) = 1$. **Thm 2.** finite irreducible + aperiodic $\Rightarrow \lim_{k \rightarrow \infty} p(k) = w$ for any $p(0)$. Irreducible alone need not converge: $P = [0, 1; 1, 0]$ oscillates, though $w = (1/2, 1/2)$. Not unique: $P = I$, every distribution is stationary. **PageRank ex.**
 $P = [1/2, 1/2; 0, 1/2; 0, 1/2; 0, 1, 0]$ is irreducible/aperiodic; $w = (2/5, 2/5, 1/5)$, check $wP = w$.

2 Value iteration & policy iteration
Setup. Finite S, A ; known $P_{x,x'}(a)$, $r(x, a) = \mathbb{E}[r(x, a)]$ (model-based). Discounted Bellman optimality:
 $V^*(x) = \max_a \mathbb{E}[r(x, a) + \gamma V^*(x')]$. **Value iteration (VI).** Start V_0 ; update $V_{k+1}(x) \leftarrow \max_a \{\bar{r}(x, u) + \gamma \sum_{x'} P_{x,x'}(a) V_k(x')\}$. Then $V_k \rightarrow V^*$. Optimal policy: $\mu^*(x) \in \arg \max_a \{\bar{r}(x, a) + \gamma \sum_{x'} P_{x,x'}(a) V^*(x')\}$.
Policy iteration (PI). Init μ_0 , iterate: (i) *eval* – solve linear Bellman eq $V_{\mu_k}(i) = \bar{r}(i, \mu_k(i)) + \gamma \sum_j P_{ij}(\mu_k(i)) V_{\mu_k}(j)$; (ii) *improve* – $\mu_{k+1}(i) \in \arg \max_a \{\bar{r}(i, u) + \gamma \sum_j P_{ij}(u) V_{\mu_k}(j)\}$; stop if $V_{\mu_{k+1}} = V_{\mu_k}$. Monotone: $V_{\mu_{k+1}} \geq V_{\mu_k}$. **Challenges.** Need $\bar{r}, P_{\mu_{k+1}}$ known; scale only to finite S, A ; otherwise must *learn*.

3 Q-learning & SARSA updates
Q-function. $Q(i, u) = \mathbb{E}[r(i, u) + \alpha V^*(i)] = \bar{r}(i, u) + \alpha \sum_j P_{ij}(u) \max_v Q(j, v)$. Optimal policy $\mu^*(i) \in \arg \max_u Q(i, u)$ (no model needed).
Q-learning (off-policy). Given experience (i, u, r, j) , TD update $\tilde{Q}(i, u) \leftarrow (1 - \beta_k) \tilde{Q}(i, u) + \beta_k (r + \alpha \max_v \tilde{Q}(j, v)) = \tilde{Q}(i, u) + \beta_k [r + \alpha \max_v \tilde{Q}(j, v) - \tilde{Q}(i, u)]$.
SARSA (on-policy). ϵ -greedy: w.p. $1 - \epsilon_k$ take $u_k \in \arg \max_u \tilde{Q}(x_k, u)$, else uniform random; $\epsilon_k \downarrow 0$. Using $(x_{k-1}, u_{k-1}, r, x_k, u_k)$: $\tilde{Q}(x_{k-1}, u_{k-1}) \leftarrow \tilde{Q}(x_{k-1}, u_{k-1}) + \beta_k [r + \alpha \tilde{Q}(x_k, u_k) - \tilde{Q}(x_{k-1}, u_{k-1})]$.
Key diff. Q-learning bootstraps *greedy* maxs $\tilde{Q}(j, v)$; SARSA bootstraps the *actually chosen* $\tilde{Q}(x_k, u_k)$.
Off- vs on-policy. Target policy = learned; behavior policy = samples. Q-learning: target = μ^* , behavior = any (each action occ-often). SARSA: target = behavior = ϵ -greedy.
Boltzmann expl. $\mu_k(x_k) = u$ w.p. $\frac{\exp(Q_k(x_k, u)/T)}{\sum_v \exp(Q_k(x_k, v)/T)}$; $T \rightarrow 0 \Rightarrow$ greedy arg max. **Ex (Q-learning).** Experience $(5, c, 0, 6)$ with $\tilde{Q}(5, c) = 0.2$, $\tilde{Q}(6, c) = -0.8$, $\tilde{Q}(6, cc) = -0.5$:
 $\tilde{Q}(5, c) \leftarrow 0.2 + \beta_k(0 + \alpha \max\{-0.8, -0.5\} - 0.2) = 0.2 + \beta_k(-0.5\alpha - 0.2)$. **Ex (SARSA).** $(4, cc, 0, 5)$; $\tilde{Q}(5, c) = 0.2 < \tilde{Q}(5, cc) = 0.3 \Rightarrow u_k = cc$.
 $\tilde{Q}(4, cc) \leftarrow 0.34 + \beta_k(0 + \alpha \cdot 0.3 - 0.34)$. **Ex (Q2 '24).** 3 states $\times$ 2 actions, $\alpha=0.9, \beta=0.1$; $Q_0 = \begin{pmatrix} 0.1 & 0.9 \\ 0.2 & 0.3 \end{pmatrix}$. Trace $(1, 2, 1) \rightarrow (2, 1, 2) \rightarrow (3, 2, 3) \rightarrow (2, 2, 0)$. Update: $Q(x_k, u_k) \leftarrow (1 - \beta) Q(x_k, u_k) + \beta [r + \alpha Q(x_{k+1}, u_{k+1})]$.
 $Q_1(1, 2) = 0.9(0.9) + 0.1(2 + 0.9(0.2)) = 0.928$.
 $Q_2(2, 1) = 0.9(0.2) + 0.1(1 + 0.9(0.5)) = 0.425$.
 $Q_3(3, 2) = 0.9(0.5) + 0.1(3 + 0.9(0.3)) = 0.777$. Other Q -entries unchanged at each step.

4 Contraction mapping & its application for proving convergence
Setup. Discounted MDP, $\alpha \in [0, 1]$.
 $V_{\mu}(i) = \mathbb{E}[\sum_{k=0}^{\infty} \alpha^k r(x_k, \mu(x_k)) \mid x_0 = i]$, $V^*(i) = \sup_{\mu} V_{\mu}(i)$.
Contraction Mapping Thm (Banach). $T: \mathbb{R}^S \rightarrow \mathbb{R}^S$ with $\|T(x) - T(y)\| \leq \alpha \|x - y\|$, $\alpha \in [0, 1]$. Then (i) $\exists x^*: x^* = T(x^*)$; (ii) $X_{k+1} = T(X_k) \rightarrow x^*$ from any X_0 .
Proof. Let $x_n = T^n(x_0)$. (a) $\|x_{n+1} - x_n\| = \|T(x_n) - T(x_{n-1})\| \leq \alpha \|x_n - x_{n-1}\| \leq \dots \leq \alpha^n \|x_1 - x_0\|$. (b) Triangle ineq.: $\|x_{n+1} - x_n\| \leq \sum_{k=n}^{\infty} \alpha^{k+1} \|x_{k+1} - x_k\| \leq (\alpha^n + \alpha^{n+1} + \dots) \|x_1 - x_0\| \leq \frac{\alpha^n}{1-\alpha} \|x_1 - x_0\| \rightarrow 0$. So $\{x_n\}$ is Cauchy; by completeness of $\mathbb{R}^S$, $\lim_{n \rightarrow \infty} x_n = x^*$ exists. (c) Fp: $\|T(x_n) - T(x^*)\| \leq \alpha \|x_n - x^*\|$, so $\lim_{n \rightarrow \infty} \alpha T(x_n) = T(x^*)$; also $T(x_n) = x_{n+1}$, so $\lim_{n \rightarrow \infty} T(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*$. Hence $T(x^*) = x^*$. (d) Unique: if

$T(y^*) = y^*$, $\|y^* - x^*\| = \|T(y^*) - T(x^*)\| \leq \alpha \|y^* - x^*\|$; since $\alpha < 1$, this forces $y^* = x^*$. **Policy eval op.**
 $T_{\mu}(V) = \bar{r}_{\mu} + \alpha P_{\mu} V$, $P_{ij} = P_{ij}(\mu(i))$;
 $T_{\mu}(x) - T_{\mu}(y) = \alpha P_{\mu}(x - y)$, so $\|T_{\mu}(x) - T_{\mu}(y)\|_{\infty} = \alpha \max_i |\sum_j P_{ij}(\mu(i) - y_j)| \leq \alpha \max_i |\sum_j P_{ij}(\mu(i) - y_j)|_{\infty} \leq \alpha \|x - y\|_{\infty}$ (rows of P_{μ} sum to 1). Hence V_{μ} unique. **Bellman optimality op.**
 $T(V(i)) = \max_u \{\bar{r}(i, u) + \alpha \sum_j P_{ij}(u) V(j)\}$; VI: $V_{k+1} = T(V_k)$. **Thm 1 (Bellman eq. for V^*).** $V^* = T(V^*)$, i.e.
 $V^*(i) = \max_u \mathbb{E}[r(i, u) + \alpha V^*(x_1) \mid x_0=i, u_0=u]$. *Proof.* Non-stationary policy $\mu = (\mu_0, \mu_1, \dots)$, tail $\mu_1 = (\mu_1, \mu_2, \dots)$.
 $(\leq)$ $V_{\mu}(i) = \bar{r}(i, \mu_0(i)) + \alpha \sum_j P_{ij}(\mu_0(i)) V_{\mu_1}(j) \leq \bar{r} + \alpha \sum_j P V^* \leq \max_u \{\bar{r} + \alpha \sum_j P V^*\}$ (used $V_{\mu_1} \leq V^*$). Taking sup: $V^*(i) \leq T(V^*)(i)$.
 $(\geq)$ Given $\varepsilon > 0$, pick $\mu^{(j)}$ with $V_{\mu^{(j)}}(i) \geq V^*(j) - \varepsilon$. Use u_0 at $t=0$, then follow $\mu^{(j)}$ when in state j : $V^*(i) \geq \bar{r}(i, u_0) + \alpha \sum_j P_{ij}(u_0) V_{\mu^{(j)}}(j) \geq \bar{r}(i, u_0) + \alpha \sum_j P_{ij}(u_0) V^*(j) - \alpha \varepsilon \forall u_0$. Take $\max_{u_0}$ then $\varepsilon \rightarrow 0$: $V^*(i) \geq T(V^*)(i)$. **Thm 2 (T is α -contraction)** (norm- ∞). *Fact 1* (monotone): if $V_1 \leq V_2$, then $T(V_1)(i) = \max_u \{\bar{r} + \alpha \sum_j P_{ij} V_1\} \leq \max_u \{\bar{r} + \alpha \sum_j P_{ij} V_2\} = T(V_2)(i)$. *Fact 2* (shift): $T(V + \gamma e)(i) = \max_u \{\bar{r} + \alpha \sum_j P_{ij}(V_j + \gamma)\} = T(V)(i) + \alpha \gamma$ (since $\sum_j P_{ij} = 1$). Let $b = \|V_1 - V_2\|_{\infty}$, so $V_2 - b e \leq V_1 \leq V_2 + b e$. Apply T (Fact 1): $T(V_2 - b e) \leq T(V_1) \leq T(V_2 + b e)$; shift (Fact 2): $T(V_2) - \alpha b e \leq T(V_1) \leq T(V_2) + \alpha b e$, so entrywise $|T(V_1) - T(V_2)| \leq \alpha b e$, hence $\|T(V_1) - T(V_2)\|_{\infty} \leq \alpha b$. Thus V^* unique, $\lim_{k \rightarrow \infty} V_k = V^*$ from any V_0 .

Thm 3 (optimal policy).
 $\mu^*(i) \in \arg \max_u \{\bar{r}(i, u) + \alpha \sum_j P_{ij}(u) V^*(j)\}$. *Proof.* By construction $T_{\mu^*}(V^*)(i) = \bar{r}(i, \mu^*(i)) + \alpha \sum_j P_{ij}(\mu^*(i)) V^*(j) = \max_u \{\dots\} = V^*(i)$ (Thm 1). Since T_{μ^*} has unique fp, $V_{\mu^*} = V^*$; so μ^* *optimal*.
5 Policy improvement & max-inequality (HW4)
PI improvement thm. With $\mu_{k+1}(i) \in \arg \max_u \{\bar{r}(i, u) + \alpha \sum_j P_{ij}(u) V_{\mu_k}(j)\}$, one has $V_{\mu_k} \leq V_{\mu_{k+1}}$. *Proof.* Define $Q_{\mu_k}(i, u) = \bar{r}(i, u) + \alpha \sum_j P_{ij}(u) V_{\mu_k}(j)$. By the update, $V_{\mu_k}(i) \leq Q_{\mu_k}(i, \mu_{k+1}(i)) = \bar{r}(i, \mu_{k+1}(i)) + \alpha \sum_j P_{ij}(\mu_{k+1}(i)) V_{\mu_k}(j)$. Plug $V_{\mu_k}(j) \leq Q_{\mu_k}(j, \mu_{k+1}(j))$ into the RHS: $V_{\mu_k}(i) \leq \mathbb{E}[r(i, \mu_{k+1}(i)) + \alpha r(j, \mu_{k+1}(j)) + \alpha^2 \sum_j P_{j,j'}(\mu_{k+1}(j)) V_{\mu_k}(j')]$. Unroll: $V_{\mu_k}(i) \leq \mathbb{E}[\sum_{t=0}^{\infty} \alpha^t r(i_t, \mu_{k+1}(i_t)) \mid i_0=i] = V_{\mu_{k+1}}(i)$.
PI convergence. If $\mu_k = \mu_{k+1}$, then μ_k is optimal. *Proof.* Def of optimality: $V_{\mu_{k+1}}(i) = \max_u \{\bar{r}(i, u) + \alpha \sum_j P_{ij}(u) V_{\mu_k}(j)\}$. Since $V_{\mu_k} = V_{\mu_{k+1}}$, this is the Bellman optimality eq.; by uniqueness $V_{\mu_k} = V^*$. **Max-inequality.**
 $|\max_x f(x) - \max_x g(x)| \leq \max_x |f(x) - g(x)|$. *Proof.* Let $x_1 \in \arg \max f$, $x_2 \in \arg \max g$; then $|\max f - \max g| = |f(x_1) - g(x_2)|$. *Case 1* $f(x_1) \geq g(x_2)$: $f(x_1) - g(x_2) \leq f(x_1) - g(x_1) \leq |f(x_1) - g(x_1)| \leq \max_x |f - g|$ (used $g(x_2) \geq g(x_1)$). *Case 2* $f(x_1) < g(x_2)$: $g(x_2) - f(x_1) \leq g(x_2) - f(x_2) \leq |g(x_2) - f(x_2)| \leq \max_x |f - g|$ (used $f(x_1) \geq f(x_2)$).

6 Update of Q-learning with linear function approximation
Linear FA. $Q(i, u) \approx \tilde{Q}(i, u; w) = w^T \phi(i, u) = \sum_{m=1}^M w_m \phi_m(i, u)$; $\phi \in \mathbb{R}^M$ given features, $w \in \mathbb{R}^M$ shared weights. **TD error.** $\delta_k = r(x_k, u_k) + \alpha \max_v w^T \phi(x_{k+1}, v) - w^T \phi(x_k, u_k)$.
Derivation via SGD. Tabular: $\tilde{Q}(x_k, u_k) \leftarrow \tilde{Q}(x_k, u_k) - \frac{1}{2} \beta_k \partial \delta_k^2 / \partial \tilde{Q}(x_k, u_k) = \tilde{Q}(x_k, u_k) + \beta_k \delta_k$. Linear FA (gradient through $w^T \phi(x_k, u_k)$, ignore max term):
 $w \leftarrow w - \frac{1}{2} \beta_k \partial \delta_k^2 / \partial w = w + \beta_k \delta_k w \phi(x_k, u_k)$.

7 Double estimator & update of DDQN
Overestimation bias. Jensen: $\mathbb{E}[\max_v X_v] \geq \max_v \mathbb{E}[X_v]$. Model noise: $\tilde{Q}(i, u) = Q(i, u) + Y_{i,u}$ with $\mathbb{E}[Y_{i,u}] = 0$. Q-learning target bias $Z = \alpha(\max_v \tilde{Q}(j, v) - \max_v Q(j, v))$ has $\mathbb{E}[Z] > 0$ in general. **Ex (iid uniform noise).** If $Q(j, \cdot)$ is constant and $Y_{j,v} \sim \text{Unif}[-\kappa, \kappa]$ iid over n actions, $\mathbb{E}[Z] = \alpha \mathbb{E}[\max_v Y_{j,v}]$. With $f = \frac{1}{2\kappa}$, $N(f) = \frac{1}{2} + \frac{\pi}{2\kappa}$, max-pdf $n f N f^{n-1}$; sub $y = \frac{1}{2} + \frac{\pi}{2\kappa}$: $\mathbb{E}[\max Y] = \kappa \int_0^1 (2y-1) y^{n-1} dy = \kappa \frac{n-1}{n+1}$. So $\mathbb{E}[Z] = \alpha \kappa \frac{n-1}{n+1}$ (grows with n).
Double Q-learning (van Hasselt '10). Maintain two nets w, w' ; at each step randomly pick one to update. Est. 1: $w \leftarrow w + \beta_k \delta_k \nabla_w Q(x_k, u_k; w)$, $\delta_k = r(x_k, u_k) + \alpha Q(x_{k+1}, u_k; w) - Q(x_{k+1}, v; w)$; $w' \leftarrow Q(x_{k+1}, u_k; w)$. Est. 2: $w' \leftarrow w' + \beta_k \delta'_k \nabla_{w'} Q(x_k, u_k; w')$, $\delta'_k = r(x_k, u_k) + \alpha Q(x_{k+1}, u_k; w') - Q(x_{k+1}, v; w')$; $w \leftarrow Q(x_{k+1}, u_k; w')$. *Select* action with one net, *evaluate* with the other.
Double estimator (underestimator) thm. For RVs

$X_1, \dots, X_M$ with sample set S_i , split $S_i = S_i^{(1)} \cup S_i^{(2)}$; empirical means $\mu_i^{(a)}$. Let $i^* \in \arg \max_i \mu_i^{(1)}$; estimate $\max_i \mathbb{E}[X_i]$ by $\mu_{i^*}^{(2)}$. *Thm:* $\mathbb{E}[\mu_{i^*}^{(2)}] \leq \max_i \mathbb{E}[X_i]$. *Proof:* if i^* is the true maximizer, $\mathbb{E}[\mu_{i^*}^{(2)}] = \mathbb{E}[X_{i^*}] = \max_i \mathbb{E}[X_i]$; otherwise $< \max_i \mathbb{E}[X_i]$. **AMSE (Weng et al '20).** With rates c/k vs $2c/(k \log k)$ (Double-Q): $\text{AMSE}(w^{(1)}) \geq \text{AMSE}(w)$; $\text{AMSE}(\frac{w^{(1)} + w^{(2)}}{2}) = \text{AMSE}(w)$. Double-Q learns faster without sacrificing AMSE.
DDQN (Double DDQN, van Hasselt-Guez-Silver). Decouple action selection from evaluation: $y_{k,b} = r_{k,b} + \alpha Q(x_{k+1,b}, \arg \max_u Q(x_{k+1,b}, u; w)$; w_{tgt} . Online net w picks action; target net w_{tgt} evaluates. Target update (periodic copy) unchanged. Cuts DDQN overestimation.
8 Score function definition & policy gradient theorem
Parameterized policy. $\pi_w(u|x) = \Pr(\text{action}=u \mid \text{state}=x)$. **Gibbs (softmax).** Finite A : $\pi_w(a|x) = \frac{\exp(w \cdot \phi(x,a))}{\sum_b \exp(w \cdot \phi(x,b))}$.
Gaussian. Cont. A : $\pi_w(a|x) = \mathcal{N}(a; \mu_w(x), \sigma^2 I)$, $\mu_w(x) = \phi(x)^T w$. **Score function.** $\nabla_w \log \pi_w(u|x)$. Gibbs: $\nabla_w \log \pi_w(u|x) = \phi(x, u) - \mathbb{E}_{u' \sim \pi_w(\cdot|x)}[\phi(x, u')]$. Gaussian: $\nabla_w \log \pi_w(u|x) = (u - \mu_w(x)) \phi(x) / \sigma^2$.
Value & PG update. $V_w(x_0) = \sum_u Q_w(x_0, u) \pi_w(u|x_0) = \mathbb{E}_{\pi_w}[\sum_{k=0}^{\infty} \alpha^k r(x_k, u_k) \mid x_0]$; $w \leftarrow w + \beta \nabla_w V_w(x_0)$. Here $\mathbb{E}_{\pi_w}[\cdot \mid x_0]$ is over the trajectory law $u_k \sim \pi_w(\cdot \mid x_k)$, $x_{k+1} \sim P(\cdot \mid x_k, u_k)$. **Policy Gradient Thm.** $\nabla_w V_w(x_0) = \mathbb{E}_{\pi_w}[\sum_{k=0}^{\infty} \alpha^k \nabla_w \log \pi_w(u_k \mid x_k) Q_w(x_k, u_k) \mid x_0]$, or equivalently = $\frac{1}{1-\alpha} \mathbb{E}_{x_0 \sim \pi_w, u \sim \pi_w(\cdot|x)}[\nabla_w \log \pi_w(u|x) Q_w(x, u)]$, where $\eta_w(x) = (1 - \alpha) \sum_{u_0} \alpha^k \Pr_{\pi_w}(x_k = x \mid x_0)$ is the *discounted state distribution*. *Proof.* Let $x_0=i$; $V_w(i) = \sum_u Q_w(i, u) \pi_w(u|i)$, so $\nabla V_w(i) = \sum_u [\nabla_w Q_w(i, u) \pi_w(u|i) + Q_w(i, u) \nabla_w \pi_w(u|i)]$. Let $\mathbb{E}_i[\cdot] := \mathbb{E}_{u_0 \sim \pi_w(\cdot|i), x_1 \sim P(\cdot|i, u_0)}[\cdot]$. *Term 1:* $Q_w(i, u) = \bar{r}(i, u) + \alpha \sum_j P_{ij}(u) V_w(j)$, so $\nabla_w Q_w(i, u) = \alpha \sum_j P_{ij}(u) \nabla_w V_w(j)$; averaging over $u_0 \sim \pi_w(\cdot|i)$ gives $\alpha \mathbb{E}_i[\nabla_w V_w(j)]$. *Term 2:* log-trick $\nabla_w \pi_w = \pi_w \nabla_w \log \pi_w$ gives $\mathbb{E}_i[Q_w(i, u_0) \nabla_w \log \pi_w(u_0|i)]$. So $\nabla V_w(i) = \mathbb{E}_i[Q_w(i, u_0) \nabla_w \log \pi_w(u_0|i)] + \alpha \mathbb{E}_i[\nabla_w V_w(j)]$; unroll to the sum: $\sum_k \alpha^k \sum_u \Pr(x_k=x \mid x_0) \pi_w(u|x) \nabla_w \log \pi_w(u|x) Q_w(x, u)$. Bundle $\sum_k \alpha^k \Pr(x_k=x \mid x_0) = \eta_w(x) / (1 - \alpha)$ to get $\frac{1}{1-\alpha} \mathbb{E}_{x_0 \sim \pi_w, u \sim \pi_w(\cdot|x)}[\nabla_w \log \pi_w(u|x) Q_w(x, u)]$.

9 REINFORCE & variance reduction
REINFORCE (Williams '88). Generate episode $x_0, u_0, \dots, x_T$ under π_w . Monte-Carlo return $Q_w(x_k) = \sum_{t=k}^T \alpha^t \alpha^{\tau-k} r(x_t, u_t)$. For each k : $w \leftarrow w + \beta \alpha^k \nabla_w \log \pi_w(u_k \mid x_k) \bar{Q}_w(x_k, u_k)$. High variance from single-episode returns. **Control variates.** To estimate $\mathbb{E}[F]$, use $F' = F - w$ with $\mathbb{E}[w] = 0$. Then $\mathbb{E}[F'] = \mathbb{E}[F]$ (unbiased); $\text{Var}(F') = \text{Var}(F) - 2\text{Cov}(F, w) + \text{Var}(w)$, smaller than $\text{Var}(F)$ when w is strongly correlated with F . **Baseline thm.** For any state-only $G_w(x)$ (action-independent): $\mathbb{E}_{u \sim \pi_w(\cdot|x)}[\nabla_w \log \pi_w(u_k \mid x_k) G_w(x_k)] = 0$. *Proof.* $\sum_u \pi_w(u|x_k) \nabla_w \log \pi_w(u|x_k) G_w(x_k) = G_w(x_k) \sum_u \pi_w(u|x_k) \nabla_w \log \pi_w(u|x_k) = G_w(x_k) \nabla_w(1) = 0$. Hence PGT holds with Q_w replaced by advantage $A_w = Q_w - G_w$: $\nabla_w V_w(x_0) = \mathbb{E}_{\pi_w}[\sum_{k=0}^{\infty} \alpha^k \nabla_w \log \pi_w(u_k \mid x_k) A_w(x_k, u_k) \mid x_0]$.
Advantage actor-critic. Pick $G_w(x) = V_w(x)$; advantage $A_w(s, a) = Q_w(s, a) - V_w(s)$. **TD residual** is an unbiased estimator of A_w : $A_w(x_k, u_k) \approx r(x_k, u_k) + \alpha V_w(x_{k+1}) - V_w(x_k)$ (same form as Q-learning TD error with V in place of w). PG estimator (let $\delta_k = r(x_k, u_k) + \alpha V_w(x_{k+1}) - V_w(x_k)$): $\nabla_w V_w(x_0) \approx \sum_{k=0}^{\infty} \alpha^k \nabla_w \log \pi_w(u_k \mid x_k) \delta_k$. Critic learns V_w (e.g. via TD(0)); actor updates with this lower-variance signal. **Ex (baseline REINFORCE).** Gaussian actor $\pi_w(u|x) = \mathcal{N}(w, x, 1)$, score $(u - wx)$. Fixed baseline $V_b(x) = x$; $w=1$, $\alpha=0.5$. Episode (x_k, u_k, r_k) : $(1, 2, 1) \rightarrow (2, 1, 2) \rightarrow (1, 2, 1)$. *MC returns:* $\hat{Q}_0 = 1+0.5(2) + 0.25(-1) = 1.75$; $\hat{Q}_1 = 2+0.5(-1) = 1.5$; $\hat{Q}_2 = -1$. *Advantages* $\hat{A}_k = \hat{Q}_k - V_b(x_k)$: $\hat{A}_0=0.75$, $\hat{A}_1=-0.5$, $\hat{A}_2=-2$. *Scores* $(u_k - wx_k) x_k$: 1, -2, 1. *Sum* $\sum_k \alpha^k \text{score}_k \hat{A}_k$: $1(1)(0.75) + 0.5(-2) - (0.5) + 0.25(1) = -2$. $\hat{w} = 0.75 + 0.5 - 0.5 = 0.75$.

10 Natural policy gradient & its connection to soft policy iteration
PG as optimization. Standard PG solves $\max_{\tilde{w}} V_w(x_0) + \nabla V_w(x_0)^T (\tilde{w} - w) - \frac{1}{2\beta} \|\tilde{w} - w\|^2$; optimum $\tilde{w} = w + \beta \nabla V_w(x_0)$. Issue: Euclidean distance in weight space doesn't reflect policy change.

NPG idea (KL proximal). Replace Euclidean by KL in policy space: $\max_{\tilde{w}} V_w(x_0) + \nabla V_w(x_0)^T (\tilde{w} - w) - \frac{1}{\beta} \mathbb{E}_{x_0 \sim \eta_w} [\log \pi_w(\cdot|x) \mid \pi_w(\cdot|x)]$. **Taylor of KL at $\tilde{w}=w$:** zeroth $\text{KL} = 0$; first $\nabla \tilde{w} \text{KL} \mid_{\tilde{w}=w} = 0$; second $\nabla^2 \tilde{w} \text{KL} \mid_{\tilde{w}=w} = \mathbb{E}_{u \sim \pi_w(\cdot|x)}[\nabla_w \log \pi_w(u|x) \nabla_w \log \pi_w(u|x)^T] = \text{FI}_w(x)$ (Fisher info). Average $\text{FI}_w = \mathbb{E}_{x \sim \eta_w} [\text{FI}_w(x)]$. **NPG update.** $w \leftarrow w + \beta \text{FI}_w^{-1} \nabla_w V_w(x_0)$; $\dagger$ = Moore-Penrose pseudo-inverse. **Convergence (tabular).** PG (Mei '20): $V^*(d) - V_{w_t}(d) \leq \frac{16S}{(1-\alpha)^2} \|\pi^* / d\|_2^2 \frac{e^2}{t}$. NPG (Agarwal '20): $V^*(x) - V_{w_t}(x) \leq \frac{1}{(1-\alpha)^2 t} \nabla x$ (rate independent of S).
Softmax (tabular). $\pi_w(u|x) = e^{w_{x,u}} / \sum_v e^{w_{x,v}}$. **Soft policy iteration.** $\pi_{t+1}(u|x) \propto \exp(\beta Q \pi_t, x, u)$. **NPG + entropy reg.** $\Leftrightarrow$ Soft PI. Entropy-regularized value: $V_{w,\tau}(x_0) = V_w(x_0) + \tau \mathbb{E}_{x \sim \eta_w} [\sum_u \pi_w(u|x) \log \frac{\pi_{k+1}(u|i)}{\pi_k(u|i)}]$. NPG on $V_{w,\tau}$: $w \leftarrow w + \beta \text{FI}_w^{-1} \nabla_w V_{w,\tau}(x_0)$ is equivalent to $\pi_{t+1}(u|x) \propto (\pi_t(u|x))^{1-\beta\tau/(1-\alpha)} \exp(\frac{\beta}{1-\alpha} Q \pi_t, x, u)$. Choose $\beta = (1 - \alpha)/\tau$: π_t -exponent $\rightarrow 0$, giving $\pi_{t+1} \propto \exp(Q \pi_t, \tau) \propto$ Soft PI. Soft PI $\rightarrow$ standard PI as $\tau \rightarrow 0$.
11 HW9
Soft PI derivation (KL-regularized improvement). Stochastic policy $\pi_k(u|i) > 0$. Step 1: evaluate $Q \pi_k$. Step 2: solve $\max_{\pi_{k+1}} \sum_i u_i \pi_{k+1}(u|i) Q \pi_k(i, u) - \beta \sum_i u_i \pi_{k+1}(u|i) \log \frac{\pi_{k+1}(u|i)}{\pi_k(u|i)}$ s.t. $\pi_{k+1}(u|i) \geq 0$, $\sum_u \pi_{k+1}(u|i) = 1$. **Claim.** $\pi_{k+1}(u|i) \propto \pi_k(u|i) \exp(Q \pi_k(i, u)/\beta)$. *Proof.* Objective separates across i . Fix i ; let $p(u) = \pi_{k+1}(u|i)$, $q(u) = \pi_k(u|i)$, $Q(u) = Q \pi_k(i, u)$, so $\max_p \sum_u p(u) Q(u) - \beta \sum_u p(u) \log \frac{p(u)}{q(u)}$ s.t. $\sum_u p(u) = 1$. Lagrangian $L(p, \lambda) = \sum_u p(u) Q(u) - \beta \sum_u p(u) \log \frac{p(u)}{q(u)} + \lambda (\sum_u p(u) - 1)$. Using $\frac{d}{dp} [p \log \frac{p}{q}] = \log \frac{p}{q} + 1$, set $\partial L / \partial p(u) = Q(u) - \beta (\log \frac{p(u)}{q(u)} + 1) + \lambda = 0$. Rearrange: $\log \frac{p(u)}{q(u)} = \frac{Q(u)}{\beta} + \frac{\lambda - \beta}{\beta}$, so $p(u) = q(u) \exp(Q(u)/\beta) \cdot \exp(\frac{\lambda - \beta}{\beta})$. Second factor is u -independent, giving $p(u) \propto q(u) \exp(Q(u)/\beta)$. Normalize: $\pi_{k+1}(u|i) = \frac{\pi_k(u|i) \exp(Q \pi_k(i, u)/\beta)}{\sum_u \pi_k(u'|i) \exp(Q \pi_k(i, u')/\beta)}$. $p(u) \geq 0$ holds automatically ($q > 0$, $\exp > 0$).

12 Midterm
Deterministic Policy Gradient (DPG). Continuous A with deterministic actor $\mu_{\theta}(x)$. Target: $\nabla_{\theta} Q(x_0, \mu_{\theta}(x_0)) (= \nabla_{\theta} V^{\mu_{\theta}}(x_0))$, since $V^{\mu_{\theta}}(x_0) = Q^{\mu_{\theta}}(x_0, \mu_{\theta}(x_0))$. Here $Q^{\mu_{\theta}}$ depends on θ through all future actions along the trajectory. Bellman recursion: $\nabla_{\theta} V^{\mu_{\theta}}(x_0) = \nabla_{\theta} Q^{\mu_{\theta}}(x_0, u) \mid_{\mu_{\theta}(x_0)} \nabla_{\theta} \mu_{\theta}(x_0) + \alpha \mathbb{E}_{x_1} [\nabla_{\theta} V^{\mu_{\theta}}(x_1)]$, which unrolls to a trajectory sum. In practice approximate $Q^{\mu_{\theta}}$ by the critic Q_w : $\nabla_{\theta} Q(x_0, \mu_{\theta}(x_0)) = \mathbb{E}_{x_0 \sim \mu_{\theta}} [\sum_{k=0}^{\infty} \alpha^k \nabla_u Q_w(x_k, u) \mid_{\mu_{\theta}(x_0)} \nabla_{\theta} \mu_{\theta}(x_k)]$.
Ex (Q4 '24). Actor $\mu_{\theta}(x) = \frac{1}{1+e^{-\theta x}}$, critic $Q_w(x, u) = (\frac{1}{2}(x+u) - 1)^2$; $\theta=5, w=1$, $\alpha=0.9$, $e \approx 3$; traj $x_0=2$, $x_1=4$, $x_2=2$. *Gradients:* $\nabla_u Q_w = \frac{w}{2}(x+u) - w \xrightarrow{w=1} \frac{1}{2}(x+u) - 1$; $\nabla_{\theta} \mu_{\theta}(x) = \frac{x e^{-\theta x}}{($

$\frac{1-\alpha}{1-\alpha}\mathbb{E}_{x\sim q_{\pi_{\theta_0}},u\sim\pi_{\theta_0}}[A_w(x,u)]$. **Applications.** (i) *TRPO* (2010): approx: if $\pi_{\tilde{\theta}}\approx\pi_w$, replace $q_w\rightarrow q_w$: $V_{\tilde{\theta}}(x_0)\approx V_w(x_0)+\frac{1}{1-\alpha}\mathbb{E}_{x\sim q_w,u\sim\pi_{\theta_0}}[A_w(x,u)]=V_w(x_0)+\frac{1}{1-\alpha}\mathbb{E}_{x\sim q_w,u\sim\pi_{\theta_0}}[\frac{\pi_{\tilde{\theta}}(u|x)}{\pi_w(u|x)}A_w(x,u)]$ (importance sampling). (ii) *Monotonic improvement*: any $\pi_{\tilde{\theta}}$ making RHS >0 strictly improves value. (iii) *TRPO opt*: $\max_u\mathbb{E}_{x\sim q_{\theta_0}}[\sum_u\pi_{\tilde{\theta}}(u|x)A_w(x,u)]$ s.t. $\mathbb{E}_{x\sim q_{\theta_0}}[KL(\pi_{\tilde{\theta}}\| \pi_w)]\leq\delta$.

14 UCB algorithm for multi-armed bandits
Setup. I arms, $X_i(t)\in[0,1]$ i.i.d. mean μ_i ; regret $R_K=\mu_1K-\mathbb{E}[\sum_kX_{i_k}(k)]$ (arm 1 optimal). ϵ -greedy: $R_K=O(K)$. **UCB.** Play each arm once; $k>1$: $i_k\in\arg\max_i[\hat{\mu}_i(k)+\sqrt{4\log K/N_i(k)}]$ (bonus = confidence interval, optimism in face of uncertainty). $R_K=O(\sqrt{K\log K})$; instance-dep: $R_K\leq(\sum_{i\geq 2}16/\Delta_i)\log K+o(1)$, $\Delta_i=\mu_1-\mu_i$. **Proof.** $B_k=\sqrt{4\log K/N_{i_k}(k)}$; $R_K=\sum_k\mathbb{E}[F_1+Y_2]$, $F_1=\mu_1-\hat{\mu}_{i_k}-B_k$, $F_2=\hat{\mu}_{i_k}+B_k-\mu_{i_k}$. *Azuma-Hoeffding*: $Y_n=\sum_{\tau\leq n}X_{i_\tau}-n\mu_i$ martingale, $|Y_{n+1}-Y_n|\leq 1\Rightarrow\Pr(|\hat{\mu}_{i,n}-\mu_i|\geq\sqrt{4\log K/n})\leq 2/K^2$. Union bound: $\Pr(A)\geq 1-2I/K$ for $A=\{|\hat{\mu}_{i,n}-\mu_i|\leq\sqrt{4\log K/n}\forall i,n\}$. On A : $F_1\leq 0$ (UCB rule $\Rightarrow\hat{\mu}_{i_k}+B_k\geq\hat{\mu}_1+B_1\geq\mu_1$), $|F_2|\leq 2B_k$; else $F_1\leq 1$, $F_2\leq 1+2\sqrt{\log K}$. Using $\sum_k1/\sqrt{N_{i_k}(k)}\leq 2I\sqrt{K}$, $\Pr(A^c)\leq 2I/K\Rightarrow R_K=O(\sqrt{K\log K})$. **Ex (Q3 '24).**

Bernoulli 2-arm, bonus $\sqrt{4\log K/N_i}$ (log = log₂), $K=1024$, log $K=10$. At $k=411$: $N_1=250$, $N_2=160$, $\hat{\mu}_1=0.6$, $\hat{\mu}_2=0.4$. (1) *Pull*: $UCB_1=0.6+\sqrt{40/250}=0.6+0.4=1.0$; $UCB_2=0.4+\sqrt{40/160}=0.4+0.5=0.9$. Pull arm 1. (2) *Update after reward* 0: $\hat{\mu}_1\leftarrow(250(0.6)+0)/251=0.5976$; $\hat{\mu}_2$ unchanged. (3) *Regret*: true $\mu_1=0.55$, $\mu_2=0.45$; pulls $1,1,2,2,1,1,2,1,2,1\Rightarrow N_1=6$, $N_2=4$. $R_K=10(0.55)-[6(0.55)+4(0.45)]=N_2\Delta_2=4(0.1)=\mathbf{0.4}$.

15 MCTS for Alpha-Zero
Architecture. Shared-backbone net outputs $(p_{\theta}(s),v_{\theta}(s))$: policy prior $p_{\theta}\in\Delta^A$ and value $v_{\theta}\in[-1,1]$. **Training samples.** Self-play produces $(s_{h,k},\pi_{h,k},z_k)$: state, MCTS-improved policy, final outcome $\in\{-1,0,+1\}$. **Loss.** $L=\sum_b(v_{\theta}(s_{h,k})-z_k)^2-\pi_{h,k}\cdot\log p_{\theta}(s_{h,k})+c\|\theta\|_2^2$ (value MSE + policy cross-entropy + L2). **MCTS simulation (4 steps)**. 1. *Selection* (PUCT): while not leaf, $a=\arg\max_u[Q(s,u)+h\,p_{\theta}(s,u)\sqrt{\sum_vN(s,v)/(1+N(s,u))}]$. 2. *Expansion*: stop at new state s'' or terminal. 3. *Add node*: new s'' added; init $N(s'',a)=Q(s'',a)=0\forall a$. 4. *Backup*. Leaf value v : if terminal, $v=+1/-1/0$ (win/lose/draw, from the moving player's view at that node; *signs alternate* along the path in 2-player games); if non-terminal new node, $v=v_{\theta}(s'')$. Along path, update Q *first* (uses old N) then N : $Q(s,a)\leftarrow(N(s,a)+Q(s,a)+v)/(N(s,a)+1)$; then $N(s,a)\leftarrow N(s,a)+1$. **Action selection.** After many sims: $\pi(a|s)=N(s,a)/\sum_aN(s,a)$. **Ex (Q5 '24).** 2-player sim $(s_1,a_1)\Rightarrow(s_2,a_2)\Rightarrow s_3$ (terminal), P2 wins ($r=\pm 1$). Before: $N(s_1,a_1)=12$, $Q=-0.5$; $N(s_2,a_2)=4$, $Q=0.8$. (1) *Backup*. Signs alternate (value from current player's view): $v_{s_2}=+1$ (P2 won), $v_{s_1}=-1$ (P1 lost). $(s_2,a_2):Q\leftarrow(4(0.8)+1)/5=0.84$, $N\leftarrow5$. $(s_1,a_1):Q\leftarrow(12(-0.5)+(1))/13=-7/13\approx-0.538$, $N\leftarrow13$. (2) *Visit-count policy*. $N(s_1,\cdot)=(13,14,10,8,5)$, sum = 50; $\pi(\cdot|s_1)=(0.26,0.28,0.20,0.16,0.10)$.

16 Potential-based reward shaping
Reward shaping. $\bar{M}=(S,A,\alpha,\bar{R})$ with $\bar{R}=R+F$; bounded $F:S\times A\times S\rightarrow\mathbb{R}$. Goal: *policy improvement*. **Naive bonus fails** (e.g. “+0.1 right”): agent cycles to farm F instead of solving task. **PBRS (Ng-Harada-Russell '99).** F is PBRS iff $\exists\phi:S\rightarrow\mathbb{R}$ with $F(x,a,\bar{x})=\alpha\phi(x')-\phi(x)$ (action-indep., telescoping). **Thm.** Suff: PBRS $\Rightarrow M$ and $\bar{M}$ share optimal policies. *Nec*: non-PBRS $F\nRightarrow\exists P,R$ with optimal policies not coincident. *Proof (suff)*. From $Q^*(x,a)=\mathbb{E}[R+\alpha\max_uQ^*(x',u)]$, subtract $\phi(x)$ and add/subtract $\alpha\phi(x')$: $Q^*(x,a)-\phi(x)=\mathbb{E}[R+(\alpha\phi(x')-\phi(x))+\alpha\max_u(Q^*(x',u)-\phi(x'))]$. Set $\hat{Q}:=Q^*-\phi$; RHS becomes $\mathbb{E}[\hat{R}+\alpha\max_u\hat{Q}(x',u)]$, i.e. $\hat{Q}$ is a fp of $\bar{M}$'s Bellman operator; uniqueness $\Rightarrow\hat{Q}=\hat{Q}^*$. Then $\bar{\pi}^*=\arg\max_u(Q^*(x,u)-\phi(x))=\arg\max_uQ^*(x,u)=\pi^*$ ($\phi(x)$ drops from argmax). **Intuition.** ϕ = state potential; cyclic paths give zero net shaping (telescoping cancels), so no farming. **Ex (PBRS).** 3-state chain s_1,s_2,s_3 ; $\alpha=0.5$, potential $\phi(s_1)=0$, $\phi(s_2)=2$, $\phi(s_3)=4$. *F values*: $F(s_1,\cdot,s_2)=0.5(2)-0=1$; $F(s_2,\cdot,s_3)=0.5(4)-2=0$;

$F(s_2,\cdot,s_1)=0.5(0)-2=-2$; $F(s_3,\cdot,s_2)=0.5(4)-4=-3$. *Cycle check* $s_1\rightarrow s_2\rightarrow s_1$: discounted sum $\alpha^0(1)+\alpha^1(-2)=1-1=0$; no farming.

17 DPO & DPO loss
Bradley-Terry. $P(y_1\succ y_2|x)=\sigma(r(x,y_1)-r(x,y_2))$, $\sigma(z)=1/(1+e^{-z})$; inverse (logit) $\sigma^{-1}(p)=\log\frac{p}{1-p}$. **KL-regularized bandit (RLHF objective).** $\max_{\pi}\mathbb{E}_{x\sim d_0,y\sim\pi(\cdot|x)}[r(x,y)]-\beta\mathbb{E}_{x\sim d_0}[KL(\pi(\cdot|x)\|\pi_{\text{ref}}(\cdot|x))]$. **Closed-form optimum (soft P1 / Gibbs).** $\pi^*(y|x)=\frac{1}{Z(x)}\pi_{\text{ref}}(y|x)\exp(r(x,y)/\beta)$. *Proof sketch* (Langragian + KKT, cf. HW9). $\partial_{\pi}[\pi r-\beta\pi\log(\pi/\pi_{\text{ref}})+\lambda(\sum\pi-1)]=0\Rightarrow r(x,y)-\beta\log(\pi^*/\pi_{\text{ref}})+1+\lambda=0$; exponentiate; λ part is y -indep. $\Rightarrow$ normalize. **DPO trick: invert for r .**

$r(x,y)=\beta\log\frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)}+\beta\log Z(x)$. Substitute into BT: differences cancel $\log Z(x)$ (prompt-dependent normalizer drops):

$P(y_1\succ y_2|x)=\sigma\left(\beta\log\frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}-\beta\log\frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)}\right)$. So the optimal policy itself acts as the implicit reward — no reward model needed. **DPO loss** (cross-entropy / logistic regression on preference dataset $D_p=\{(x_k,y_{w,k},y_{l,k})\}$): $L_{\text{DPO}}(\theta)=-\mathbb{E}_{(x,y,w,l)\sim D_p}\left[\log\sigma\left(\beta\log\frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)}-\beta\log\frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]$.

Ex (DPO). $\beta=1$, one sample: $\pi_{\text{ref}}(y_w|x)=0.2$, $\pi_{\text{ref}}(y_l|x)=0.4$; $\pi_{\theta}(y_w|x)=0.4$, $\pi_{\theta}(y_l|x)=0.2$. *Ratios*: $\pi_{\theta}/\pi_{\text{ref}}(y_w)=2$, $\pi_{\theta}/\pi_{\text{ref}}(y_l)=0.5$; log-ratios $\ln 2,-\ln 2$. *σ -arg*: $z=\beta[\ln 2-(-\ln 2)]=2\ln 2=\ln 4$. *Sigmoid*: $\sigma(\ln 4)=1/(1+1/4)=4/5=0.8$. *Loss*: $L_{\text{DPO}}=-\ln 0.8=-\ln(5/4)=\mathbf{0.223}$. Sanity: if $\pi_{\theta}=\pi_{\text{ref}}$ $\Rightarrow z=0$, $\sigma=0.5$, $L=\ln 2=0.693$ (larger, as expected).

18 Zeroth-order optimization
ZO estimator (gradient-free). Want $\min_x f(x)$ when ∇f unavailable. Sample $u\sim\mathcal{N}(0,I)$; estimator $g_{\mu}(x;u)=\frac{\mu}{f(x+\mu u)-f(x)}u$. Update $x_{k+1}=x_k-h\,g_{\mu}(x_k;u_k)$. **Gaussian-smoothed** $f_{\mu}(x)=\mathbb{E}_{u\sim\mathcal{N}(0,I)}[f(x+\mu u)]$ ($f_{\mu}\rightarrow f$ as $\mu\rightarrow 0$). **Key identity.** $\nabla f_{\mu}(x)=\mathbb{E}_u[g_{\mu}(x;u)]\Rightarrow$ ZO is SGD on f_{μ} . *Proof.* Let $\phi(u)=\frac{1}{(2\pi)^{n/2}}e^{-\|u\|^2/2}$. $\nabla f_{\mu}(x)=\nabla_x\int f(x+\mu u)\phi(u)du$; sub $y=x+\mu u\Rightarrow\nabla f_{\mu}(x)=\nabla_x\int f(y)\frac{e^{-\|y-x\|^2/(2\mu^2)}}{(2\pi)^{n/2}\mu^n}dy$. Differentiating $e^{-\|y-x\|^2/(2\mu^2)}$ w.r.t. x gives factor $(y-x)/\mu^2$; re-substitute $u=(y-x)/\mu$: $\nabla f_{\mu}(x)=\mathbb{E}_u[u f(x+\mu u)/\mu]$. Since $\mathbb{E}[u]=0$, $\mathbb{E}[f(x+\mu u)]=0$; subtract: $\nabla f_{\mu}(x)=\mathbb{E}_u[\frac{f(x+\mu u)-f(x)}{\mu}u]=\mathbb{E}_u[g_{\mu}(x;u)]$. **SZPO (for RLHF).** No explicit reward — use “perturb + compare”. Iter t : (1) sample $v_t\sim\mathcal{N}(0,I)$, perturbed $\theta_t^*= \theta_t+\mu v_t$; (2) sample trajectory batches from $\pi_{\theta_t},\pi_{\theta_t^*}$; query M evaluators $\times N$ pairs; (3) compute scalar s_t (see below); direction $\hat{g}_t=s_tv_t$; (4) update $\theta_{t+1}=\theta_t+\alpha_t\hat{g}_t$. **Computing s_t .** By ZO: $s_t=[V(\theta_t^*)-V(\pi_{\theta_t})]/\mu$. *Known link* (BT): $r(\tau^*)-r(\tau)\approx\sigma^{-1}(\hat{P}(\tau^*)-\tau)$, so $s_t\approx\frac{1}{\mu\sqrt{N}}\sigma^{-1}(\hat{P}_N)$.

Unknown link (Sign-SZPO): $s_t=\text{sign}(\mathbb{E}[r(\tau^*)-r(\tau)])$ via majority vote. Both converge. **Ex (ZO, mini-batch $N=3$).** $f(x)=x_1^2+x_2^2$, $x_k=(2,1)$, $\mu=1$, $h=1$; true $\nabla f(4,2)$. Samples $u^{(1)}=(1,1)$, $u^{(2)}=(1,-1)$, $u^{(3)}=(-1,1)$; $f(x_k)=5$; $f(x_k+\mu u^{(i)})=(3,2)$, $(3,0)$, $(1,2)$ give 13,9,5. Estimators $g_{\mu}=\frac{f(x_k+\mu u^{(i)})-f(x_k)}{\mu}u^{(i)}$: (8,8), (4,-4), (0,4) — each is biased single-sample. Avg $\bar{g}=\frac{1}{3}(12,4)$, $(4,4/3)\approx(4,1.33)$; update $x_{k+1}=(2,1)-0.1(4,4/3)=\mathbf{(1.6,0.867)}$, very close to GD step $(1.6,0.8)$. **Ex (SZPO).** 2D; $s_t=(1,0)$, $v_t=(1,1)$, $\mu_1=1$, $\alpha_1=0.1$, $N=1$; humans give $\hat{P}(\tau^*\succ\tau)=0.8$. Perturbed $\theta^*=\theta_t+\mu_1v_t=(2,1)$. *BT*: $\sigma^{-1}(0.8)=\log 4\approx 1.386$; $s_t=\frac{1}{\mu_1N}\log 4=\log 4$;

$\hat{g}_t=s_tv_t=(\log 4,\log 4)$; $\theta_{t+1}=(1,0)+0.1(\log 4,\log 4)=\mathbf{(1.139,0.139)}$. *Sign*: majority prefers perturbed $\Rightarrow s_t=+1$; $\hat{g}_t=(1,1)$; $\theta_{t+1}=\mathbf{(1.1,0.1)}$. BT uses confidence magnitude (step scales with $\hat{P}$); Sign ignores magnitude, robust when link unknown. **Ex (HW9 Q2, 1D ZO convergence).** Scalar convex f , unique x^* = argmin; $u\sim\mathcal{N}(0,1)$, $g_{\mu}(x_k)=\frac{f(x_k+\mu u_{\mu})-f(x_k)}{\mu}u_k$, $x_{k+1}=x_k-hg_{\mu}$. (a) *Recurrence*: square $x_{k+1}-x^*=(x_k-x^*)-hg_{\mu}$ & take expectation:

$\mathbb{E}[(x_{k+1}-x^*)^2]=\mathbb{E}[(x_k-x^*)^2]-2h\mathbb{E}[g_{\mu}(x_k)(x_k-x^*)]+h^2\mathbb{E}[g_{\mu}^2]$. (b) *Avg-iterate bound* (assume $\mathbb{E}[g_{\mu}^2]\leq g_{\max}$): $\frac{1}{T+1}\sum_{t=0}^T\mathbb{E}[f_{\mu}(x^{(t)})-f_{\mu}(x^*)]\leq\frac{(2x_0-x^*)^2}{(T+1)h}+g_{\max}\frac{h}{2}$. *Proof*: let $\Delta_k=\mathbb{E}[(x_k-x^*)^2]$. Tower (condition on x_k ; x_k-x^* is x_k -meas, so pulls out): $\mathbb{E}[g_{\mu}(x_k)(x_k-x^*)]=\mathbb{E}[\mathbb{E}[g_{\mu}(x_k)(x_k-x^*)|x_k]]=\mathbb{E}[(x_k-x^*)\mathbb{E}[g_{\mu}|x_k]]=\mathbb{E}[(x_k-x^*)\nabla f_{\mu}(x_k)]$ (ZO unbiasedness). Convexity: $f_{\mu}(x_k)-f_{\mu}(x^*)\leq(x_k-x^*)\nabla f_{\mu}(x_k)$. Sub into (a) with $\mathbb{E}[g_{\mu}^2]\leq g_{\max}$: $\Delta_{k+1}\leq\Delta_k-2h\mathbb{E}[f_{\mu}(x_k)-f_{\mu}(x^*)]+h^2g_{\max}$. Rearrange: $2h\mathbb{E}[f_{\mu}(x_k)-f_{\mu}(x^*)]\leq\Delta_k-\Delta_{k+1}+h^2g_{\max}$. Sum $k=0..T$; Δ_k telescopes: $\sum_k(\Delta_k-\Delta_{k+1})=\Delta_0+\Delta_{T+1}\leq\Delta_0=(x_0-x^*)^2$. So $2h\sum_k\mathbb{E}[f_{\mu}(x_k)-f_{\mu}(x^*)]\leq(x_0-x^*)^2+(T+1)h^2g_{\max}$; divide by $2h(T+1)$. Optimal $h^*=|x_0-x^*|/\sqrt{(T+1)g_{\max}}$ balances RHS; rate $O(1/\sqrt{T})$. (c) *Smoothing error* (if L = L -Lipschitz): $|f_{\mu}(x)-f(x)|\leq\mu L\mathbb{E}[|u|]=\mu L\sqrt{2/\pi}$. *Proof*: $f_{\mu}(x)-f(x)=\mathbb{E}[f(x+\mu u)]-f(x)=\mathbb{E}[f(x+\mu u)-f(x)]$. Jensen $|\mathbb{E}[X]|\leq\mathbb{E}[|X|]$: $|f_{\mu}(x)-f(x)|\leq\mathbb{E}[|f(x+\mu u)-f(x)|]$. Lipschitz: $|f(x+\mu u)-f(x)|\leq L|\mu u|=\mu L|u|$, so $\mathbb{E}[|\cdots|]\leq\mu L\mathbb{E}[|u|]$. $u\sim\mathcal{N}(0,1)\Rightarrow\mathbb{E}[|u|]=\sqrt{2/\pi}$ (half-normal). Small $\mu\Rightarrow f_{\mu}\approx f$.

19 Contrastive learning
Idea. Density estimation $\rightarrow$ binary classification: positive $x^+\sim P_+$ (data), negative $x^-\sim P_-$ (noise, our choice); classifier f_{θ} with $P(x\in D)\approx\sigma(f_{\theta}(x))$. Avoids partition function. **BCE loss.** $L(\theta)=-\mathbb{E}[y\log\sigma(f_{\theta}(x))+(1-y)\log(1-\sigma(f_{\theta}(x)))]$. **Thm (Bayes-optimal classifier).** With P_+ , P_- equally likely, $f^*(x)=\log(P_+(x)/P_-(x))$ (log-density ratio); equiv. $P_+(x)\propto e^{f^*(x)}P_-(x)$. *Proof.* Write $L=-\frac{1}{2}\int\| \log\sigma(f(x))P_+(x)+\log(1-\sigma(f(x)))P_-(x)\|dx$. Pointwise $d/d f(x)=0$ uses $\sigma'(x)=1-\sigma(x)$: $P_+(1-\sigma(f^*))-P_-\sigma(f^*)=0\Rightarrow\sigma(f^*)=\frac{P_+}{P_++P_-}\Rightarrow f^*=\log(P_+/P_-)$. **Goal-conditioned RL.** No reward; goal prior $\psi(g)$; policies $\pi(a|s,g)$, random-goal marginal $\pi(a|s)=\sum_g\pi(a|s,g)\psi(g)$. Goal reward $r_g(s,a)=(1-\gamma)p(s_{t+1}=g|s,a)$; $\pi^*(s,a,g)=\mathbb{E}_{\pi}[\sum_t\gamma^t r_g(s_t,a_t)|s_0,a_0]$. **Q as occupancy.** $Q^{\pi}(s,a,g)=p^{\pi}(s_+=g|s,a)$ where s_+ is state at $\tau\leftarrow\text{Geom}(1-\gamma)$. Q is itself a distribution over g . **Contrastive RL.** Let $x=(s,a,g)$; $P_+(x)=p^{\pi}(g|s,a)d_0(s,a)$, $P_-(x)=\psi(g)d_0(s,a)$. By CL thm, $f^*(s,a,g)=\log\frac{P_+}{P_-}=\log\frac{Q^{\pi}(s,a,g)}{\psi(g)}$, so $Q^{\pi}(s,a,g)=e^{f^*(s,a,g)}\psi(g)$ and $\arg\max_a f^*=\arg\max_a Q^{\pi}$. No TD/Bellman — pure supervised classification. **Sampling.** *Positive*: $(s,a)\sim d_0$, roll out π , $g^+=s_{\tau+}$, $t^+\leftarrow\text{Geom}(1-\gamma)$, label +. *Negative*: (s,a,g^-) , $g^-=f^*$ independent, label $-$. **Param.** Log-linear $f(s,a,g)=\phi(s,a)^{\top}\psi(g)$ (two encoder nets); emergent representation learning. **Ex (MC estimate of p^{π}).** Run $N=5$ rollouts from (s_1,a_1) ; each step stop w.p. $1/3$; suppose stop states are $\{s_2,s_3,s_2,s_2,s_3\}$. Then $\hat{p}^{\pi}[s_1|s_1,a_1]=0,3/5,2/5$ for (s_1,s_2,s_3) ; converges to true p^{π} as $N\rightarrow\infty$. **Ex (CL-RL).** $S=\{s_1,s_2,s_3\}$, uniform g ; (1)=1/3; observe $p^{\pi}(s_+|s_1,a_1)=(0.5,0.3,0.2)$ over (s_1,s_2,s_3) . *Q*: $q^{\pi}(s_1,a_1,g)=p^{\pi}(s_+=g|s_1,a_1)=(0.5,0.3,0.2)$. *f*: $f^*=\log(Q^{\pi}/\psi)=\log(3Q^{\pi})=(\log 1.5,\log 0.9,\log 0.6)\approx(0.405,-0.105,-0.511)$. *Argmax*: both yield $s_1\checkmark$. *Recover*: $Q^{\pi}=e^{f^*}\psi=(1.5,0.9,0.6)/3=(0.5,0.3,0.2)\checkmark$. Sign of f^* says g is easier (+) or harder (−) to reach than prior.

20 Relative value function & Blackwell optimality
Avg-reward MDP. $\max\lim_{N\rightarrow\infty}\frac{1}{N}\sum_{k=0}^{N-1}\mathbb{E}[r(x_k,u_k)]$. Under unichain μ : $V_0^{\mu}(i)=\sum_j\pi_jr(j,\mu(j))$ — independent of i (stationary dist dominates). **Relative value.** From $V_0^{(N)}(i)=\max\sum_{k=0}^{N-1}\mathbb{E}[r|x_0=i]$, set $h_N(i):=V_0^{(N)}(i)-NV^*$; substitute & let $N\rightarrow\infty$: $V^*+h(i)=\max_u\{r(i,u)+\sum_jP_{ij}(u)h(j)\}$. V^* : optimal avg reward (scalar); $h(i)$: transient bonus relative to average, $V_0^{(N)}(i)\approx NV^*+h(i)$. Normalize $h(0)=0$ (additive freedom). **Optimality thm.** (V^*,h) solving Bellman $\Rightarrow$ argmax policy μ^* is optimal, V^* is optimal avg reward. *Proof.* $V^*\geq r(i,\mu^*)+\mathbb{E}[h(x_{k+1})|\mu]-h(i)$; take $\mathbb{E}_{x_0}$ + telescoping over $k=0..N-1$: $NV^*\geq\sum_k\mathbb{E}[r(i,\mu^*)+\mathbb{E}[h(x_N)]-\mathbb{E}[h(x_0)]]$; Let's $/N$ then $N\rightarrow\infty$ (h is bounded) $\Rightarrow V^*\geq\bar{r}^{\mu}$; equality at μ^* . **Relative V1.** $\bar{V}_1(V|\alpha=1)$ diverges ($V_k\approx kV^*+h$). Use relative values $V_k=V_k-V_k(x_0)$. Alg 1 (two sequences): init $\bar{V}_0(i)=0$; then iterate

$V_k(i)=\max_u\{r(i,u)+\sum_jP_{ij}(u)\bar{V}_k(j)\}$; $\bar{V}_{k+1}(i)=V_{k+1}(i)-V_{k+1}(x_0)$. Alg 2 (single sequence): $V_{k+1}(i)=\max_u\{r(i,u)+\sum_jP_{ij}(u)V_k(j)\}-V_k(x_0)$. **Policy iter (avg).** Fix μ_k ; *eval*: solve $V_k+h_k(i)=r(i,\mu_k(i))+\sum_jP_{ij}(\mu_k(i))h_k(j)\forall i$, $h_k(0)=0$. *Improve*: $\mu_{k+1}(i)=\arg\max_u\{r(i,u)+\sum_jP_{ij}(u)h_k(j)\}$; stop if $(V_{k+1},h_{k+1})=(V_k,h_k)$. **Improvement thm.** If $r(i,\mu(i))+\sum_jP_{ij}(\mu(i))h(j)\geq V+h(i)\forall i$, then $V^*\geq V$ (strict if strict holds at some positive-recurrent state). *Proof.* Multiply both sides by π_i^{μ} and sum over i : $\sum_i\pi_i^{\mu}r(i,\mu(i))+\sum_i\pi_i^{\mu}\sum_jP_{ij}(\mu(i))h(j)\geq V+\sum_i\pi_i^{\mu}h(i)$. By stationarity, $\sum_i\pi_i^{\mu}P_{ij}(\mu(i))=\pi_j^{\mu}\Rightarrow$ the h -terms cancel; $LHS=V^{\mu}$, giving $V^{\mu}\geq V$. **Avg Q-learning.** $Q(i,u):=r+\sum_jP_{ij}(u)h(j)-V^*$, so $h=\max_uQ$ and $Q(i,u)=r+\sum_jP_{ij}(u)\max_uQ(j,v)-V^*$. Surrogate $V^*\rightarrow\max_uQ_k(0,v)$. Let $\delta_k=r+\max_uQ_k(j,v)-Q_k(i,u)-\max_uQ_k(0,v)$; update $Q_{k+1}(i,u)=Q_k(i,u)+\beta_k\delta_k$. **Blackwell optimality.** μ^* is BO if optimal $\forall\alpha\in(0,1)$. For finite S,A , BO policies exist and are also avg-reward optimal. *Existence.* Finitely many det policies; along $\alpha_n\uparrow 1$ some μ^* repeats. If μ^* not BO, another $\bar{u}$ wins on subseq $\alpha_{n_i}\uparrow 1$; but $Q_{\alpha}(i,u)$ is rational in α , can't change sign α -often — contradiction. *BO $\Rightarrow$ avg opt.* Laurent $V_{\alpha}^*(x)=\frac{V^*}{1-\alpha}+h^{\mu}(x)+O(1-\alpha)$. BO gives $V_{\alpha}^{\mu}\leq V_{\alpha}^{\mu^*}$ as $\alpha\rightarrow 1$; leading term $\Rightarrow V^{\bar{\mu}}\leq\bar{V}^{\mu^*}$. **Ex (crowdsourcing).** Type- i arrives w.p. p_i , completes w.p. q_i , reward R_i ; state 0=idle, is=busy. Bellman: $V^*+h(0)=\sum_i p_i\max\{h(i),h(0)\}$; $V^*+h(i)=q_i(R_i+h(0))+(1-q_i)h(i)$. Set $h(0)=0\Rightarrow h(i)=R_i-V^*/q_i$; sub: $V^*=\sum_i p_i\max(R_i-V^*/q_i,0)$ (implicit in V^*). Optimal: accept type- i iff $R_i\geq V^*/q_i$. **Ex (avg-reward 2-state MDP).** States $\{0,1\}$, actions $\{s,u\}$; det. $(0,s)\rightarrow(0,w)\rightarrow 1$, $(1,s)\rightarrow 1$, $(1,w)\rightarrow 0$; rewards $r(0,s)=1$, $r(0,w)=2$, $r(1,s)=3$, $r(1,w)=0$. Ref $x_0=0$. Optimum: $V^*=3$, $h(0)=1$, $\mu^*=(w,s)$. *Relative V1* (Alg 2): $V_0=(0,0)$. $V_1(0)=\max\{1,2\}-0=2$, $V_1(1)=\max\{3,0\}-0=3$. $V_2(0)=\max\{1+2,2+3\}-2=3$, $V_2(1)=\max\{3+3,0+2\}-2=4$. Converged: $V_k\rightarrow(V^*+h)=(3,4)$. *PI (avg)*. $\mu_0=(s,w)$. *Eval 1* ($h_0(0)=0$): $V_0=1$, $h_0(1)=-1$. *Improve 1*: $i=0$: $\max\{1+0,2+(-1)\}=\max\{1,1\}$ (tie, take w); $i=1$: $\max\{2,0\}\rightarrow s$; so $\mu_1=(w,s)$. *Eval 2*: $V_1=3$, $h_1(1)=1$. *Improve 2*: $(w,s)=\mu_1$ — stop. *Avg Q-learning.* $Q_k=\binom{1}{4}^2$ (rows = state, cols = $\{s,w\}$), $\beta_k=0.5$; experience $(i=0,u=w,r=2,j=1)$: $\delta_k=2+4\max\{4,0\}-2-\max\{1,2\}=2+4-2-2=2$; $Q_{k+1}(0,w)=2+0.5(2)=3$.

21 Duality & KKT conditions
Primal. $\max_{x\in S} f(x)$ s.t. $h_i(x)\leq 0$ ($i=1..I$), $g_j(x)=0$ ($j=1..J$). **Lagrangian.** $L(x,\lambda,\mu)=f(x)-\sum_i\lambda_ih_i(x)+\sum_j\mu_jg_j(x)$, $\lambda_i\geq 0$ (μ_j unrestricted). **Dual function.** $D(\lambda,\mu):=\sup_{x\in S}L(x,\lambda,\mu)$. **Weak duality thm.** (i) D is convex in (λ,μ) ; (ii) $D(\lambda,\mu)\geq f^*$. *Proof.* (i) L is affine in (λ,μ) ; pointwise sup of affine functions is convex. (ii) For feasible x : $-\sum_i\lambda_ih_i\geq 0$ (since $\lambda_i\geq 0$, $h_i\leq 0$) and $\sum_j\mu_jg_j=0$, so $L(x,\lambda,\mu)\geq f(x)$; sup over feasible x gives $D\geq f^*$. **Dual problem.** $d^*:=\inf_{\lambda\geq 0,\mu}D(\lambda,\mu)$. *Duality gap* $d^*-f^*\geq 0$; strong duality: $d^*=f^*$. **Slater's condition** (sufficient for strong duality): f concave, h_i convex, g_j affine, and $\exists x\in\text{reint}(S)$ with $h_i(x)<0\forall i$, $g_j(x)=0\forall j$. **KKT conditions.** f,h_i diff, Slater holds. Then x^* is global max iff $\exists\lambda_i^*\geq 0,\mu_j^*$: (1) *Stationarity*:

(2)

Complementary slackness: $\lambda_i^*h_i(x^*)=0\forall i$. (Plus feasibility $h\leq 0$, $g=0$; dual feasibility $\lambda_i^*\geq 0$.) Strictly convex $f\Rightarrow x^*$ unique. *CS proof.* Strong duality: $f(x^*)=D(\lambda^*,\mu^*)\geq L(x^*,\lambda^*,\mu^*)=f(x^*)-\sum_i\lambda_i^*h_i(x^*)$ ($g=0$); so $\sum_i\lambda_i^*h_i(x^*)\geq 0$. But each term $\lambda_i^*h_i(x^*)\leq 0\Rightarrow$ all = 0. **Ex (resource alloc).** Variable $x=(x_0,x_1,\dots,x_L)\in\mathbb{R}^{L+1}$: <